

COURSE CODE: DCIT 208

**ASSIGNMENT TITLE: INDIVIDUAL ASSIGNMENT-
WEEK 4-IA1**

PERSONAL SYSTEM REQUEST-FLOW DIAGRAM

FULL NAME: DZUDZOR SELASI ANNETTE ESI

INDEX NUMBER:22310101

TEAM NAME: INNOVATEX

PROJECT NAME: SCENTHUB

TEAM REPOSITORY URL:

<https://github.com/selasi07/ScentHub>

INDIVIDUAL PORTFOLIO REPOSITORY URL:

<https://github.com/selasi07/DCIT208-Portfolio.git>

YOUTUBE LINK:

<https://youtu.be/3k8lp4Urjgw?si=pgqwDv6FEeHbztTw>

I used AI assistance. I used CHATGPT as a learning aid to better understand the assignments and clarify software engineering concepts. The guidance provided helped me structure and complete the assignment. All final work was reviewed, adapted, and submitted based on my own understanding of the ScentHub project. I understand that I must be able to explain and defend every part of this submission.

REQUEST FLOW-DIAGRAM

NAME : DZUDZOR SELASI ANNETTEESI
STUDENT ID : 22310101
TEAM : InnovateX
PROJECT : ScentHub
FEATURE : Record Sale
DATE : Thursday 25 June 2026

REQUEST FLOW
[user/sales Attendant]
↓
[Web Browser]
↓
<https://scenthub.com/sales>
↓

[DNS Lookup]
↓

HTTPS POST request (product_id, quantity, customer_name)
↓

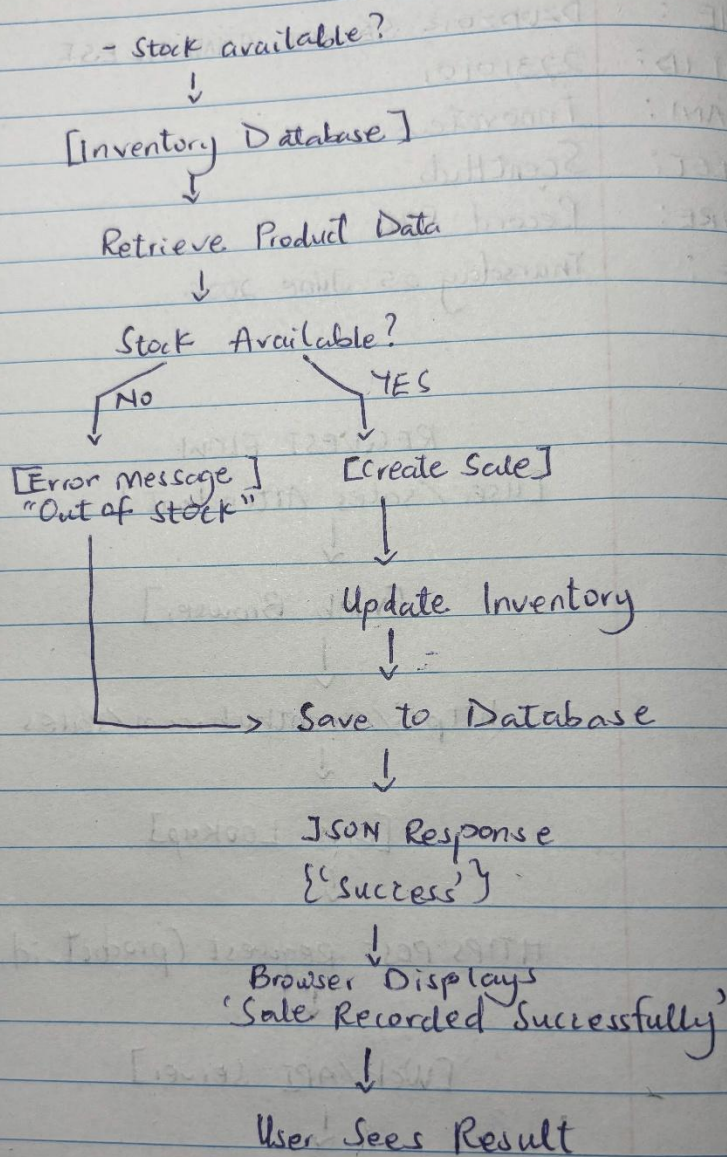
[Web/API server]
↓

[sales controller]
↓

[Input validation]

- Product exists?

- Quantity > 0?



Failure Point:

Database Connection fails during sales recording

Security Concern:

Unauthorized users accessing sales or customer data

PART B

1. WHAT DEVICE IS THE USER ON?

The user is likely a perfume shop owner or sales attendant using a laptop, desktop computer, tablet, or smartphone. They use the device to access ScentHub and record sales.

2. IS IT A BROWSER OR A MOBILE APP?

ScentHub is currently planned as a web-based system, so users will access it through a browser such as Chrome or Firefox. This allows staff to manage sales and inventory online.

3. What URL, route, screen, or endpoint is involved?

When recording a sale, the user opens the sales page and enters the transaction details. Once submitted, the information is sent to the sales endpoint, which processes and stores the sale.

4. What protocol is used?

The system uses **HTTPS** to send data securely between the browser and the server. This helps protect business information from unauthorized access.

5. What data is sent?

The user sends information such as the perfume name, quantity sold, price, total amount, date of sale, and staff details. This data is needed to record the transaction accurately.

6. What validates the request?

Before saving the sale, the system checks that all required information is provided, the product exists, enough stock is available, and the user is authorized to perform the action.

7. What database table, collection, or external service is touched?

The system updates the **Products** table to reduce stock and adds a new record to the **Sales** table to store the transaction details.

8. What response comes back?

If everything is successful, the system displays a message confirming that the sale has been recorded and updates the inventory information.

9. What happens if validation fails?

If there is missing or incorrect information, such as an invalid quantity or insufficient stock, the system displays an error message and prevents the sale from being recorded.

10. What happens if the database is down?

If the database is unavailable, the sale cannot be saved. The system informs the user of the problem and asks them to try again later.

11. What happens if 500 users do this at once?

The system may become slower because of the large number of requests. To handle this, the team can use techniques such as load balancing and database optimization to keep the system running smoothly.

12. What security or privacy concern exists?

A major concern is unauthorized access to sales and inventory data. To protect this information, ScentHub should use secure logins, strong passwords, HTTPS encryption, and role-based access control.

EXPLANATION OF THE REQUEST FLOW FOR RECORDING A SALE IN SCENTHUB

For this assignment, I selected the **Record Sale** feature from our team project, **ScentHub**. ScentHub is a system being developed to help perfume businesses manage their inventory, sales records, and business reports more efficiently. Currently, our client manages these activities manually, which can be time-consuming and may lead to mistakes. The goal of ScentHub is to provide a centralized platform where sales and inventory information can be stored, updated, and accessed easily.

The feature I chose is important because every sale made by the business affects inventory levels, sales records, and future business reports. Understanding how a sales request moves through the system helps me understand how different components of software systems work together.

Step 1: The User Initiates the Request

The request flow begins when a sales attendant or shop owner wants to record a perfume sale. The user accesses ScentHub using a device such as a laptop, desktop computer, tablet, or smartphone. Since the system is currently planned as a web application, the user interacts with it through a web browser such as Google Chrome or Firefox.

The user navigates to the sales section of the system and enters details such as the perfume being sold, the quantity purchased, and the selling price. Once all the information has been entered, the user clicks the submit button to record the sale.

Step 2: Browser and URL Processing

After the user submits the form, the browser prepares the information and sends it to the system. Before the request can reach the server, the browser uses the system's URL to determine where the request should go.

The URL acts like an address that helps the browser locate the correct server. The browser may access a page such as the sales entry screen and send the request to a specific endpoint responsible for handling sales transactions.

Step 3: DNS and Routing

When the browser receives the URL, it needs to determine the location of the server hosting ScentHub. This is where the Domain Name System (DNS) becomes important.

DNS converts the human-readable domain name into an IP address that computers can understand. Once the correct IP address has been found, the request is routed through the internet to the appropriate server.

Although users do not normally see this process, it happens automatically every time they access a website.

Step 4: HTTPS Request

The browser sends the request using HTTPS (Hypertext Transfer Protocol Secure). HTTPS is important because it protects information while it travels across the internet.

For example, sales information, inventory data, and user account details should not be visible to unauthorized individuals. HTTPS encrypts the communication between the browser and the server, making it much more secure.

This security measure helps protect both the business and its customers.

Step 5: Server Receives the Request

Once the request reaches the server, the application begins processing it. The server acts as the central component responsible for handling requests from users and performing the required actions.

At this stage, the server receives all the information entered by the sales attendant and prepares to validate it before saving it to the database.

Step 6: Validation of Data

Validation is one of the most important parts of the request flow. Before recording the sale, the system must ensure that the information provided is correct and complete.

The system checks whether all required fields have been filled in correctly. It verifies that the product exists in the inventory, confirms that the quantity entered is greater than zero, and checks whether enough stock is available for the sale. The system also ensures that the user is properly logged in and authorized to perform the action.

These checks help prevent errors and maintain accurate records within the system.

Step 7: Database Interaction

If all validation checks pass successfully, the server communicates with the database.

Two important database tables are involved in this process. The first is the Products table, which contains information about the perfumes currently in stock. The second is the Sales table, which stores records of completed sales transactions.

When a sale is recorded, a new entry is added to the Sales table. At the same time, the quantity available in the Products table is reduced to reflect the items that have been sold.

This process ensures that inventory levels remain accurate and up to date.

Step 8: Response Sent Back to the User

After the database has been updated successfully, the server generates a response and sends it back to the browser.

The response informs the user that the sale has been recorded successfully. The browser then displays a confirmation message on the screen. In some cases, the inventory display may also be updated automatically to show the new stock levels.

This allows the user to immediately see the result of their action.

Possible Failure Point

One possible failure point in this request flow is database unavailability.

If the database server is down or cannot be reached, the application will not be able to save the sales transaction. As a result, inventory records will not be updated, and the user may receive an error message indicating that the system is temporarily unavailable.

To reduce the impact of such failures, the system should include proper error handling and regular database backups.

Security and Privacy Concern

A major security concern for ScentHub is unauthorized access to business information.

Since the system stores sales records and inventory data, unauthorized users could potentially view, modify, or delete important information if proper security controls are not implemented.

To reduce this risk, the system should use HTTPS encryption, secure authentication methods, strong passwords, and role-based access controls. These measures ensure that only authorized users can access sensitive information.

Handling Large Numbers of Users

Although the perfume business is currently small, it is important to consider future growth. If hundreds of users attempt to record sales at the same time, the system could experience increased workload and slower response times.

To address this challenge, the development team can implement techniques such as database optimization, caching, load balancing, and cloud-based scaling. These approaches help ensure that the system remains responsive even during periods of high demand.

Conclusion

The Record Sale feature demonstrates how different components of a software system work together to complete a user's request. The process begins when the user enters sales information and ends when the sale is successfully stored and confirmed. Along the way, the request passes through the browser, DNS, HTTPS communication, server processing, validation checks, and database operations before a response is returned.

By studying this request flow, I gained a better understanding of how modern web applications process user requests, maintain data accuracy, handle failures, and protect sensitive information. This knowledge will help me contribute more effectively to the development of ScentHub as the project continues.

AI DISCLOSURE TABLE

Tool	Purpose	Output used	Verification
ChatGPT	Understanding assignment requirements	Guidance on assignment structure	Compared with assignment brief
ChatGPT	Explaining software engineering concepts	Explanations of request flow, validation, database, and security	Verified against lecture materials and project context
ChatGPT	Answering questions and clarifying uncertainties	Conceptual guidance only	Reviewed and adapted before use